Dialectica categories without subobjects

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Abstract

Dialectica categories are usually presented with the relation of an object (U, X, α) given as a subobject of $U \times X$. It has long been known that they can equally be presented with the relation given as a map from $U \times X$ into a lineale, a poset with a commutative monoidal structure, monotone in each argument, and an implication right adjoint to it. The two-element lineale over sets gives back the subobject presentation. For the Girard variant of the dialectica construction the calculations in this form are written down, but for the original construction Dial($\mathcal{C}$) they are not. This note records them. With the lineale an internal poset in a cartesian closed base with finite coproducts, Dial($\mathcal{C}$) is symmetric monoidal closed, with finite products and weak coproducts. The adjunction between the tensor and the linear implication in Dial($\mathcal{C}$) is given by the corresponding adjunction in the lineale. The modalities also work as expected.

1 Introduction

Dialectica categories were introduced in [1] as an internal version of Gödel's Dialectica interpretation [2], and turned out to be models of Girard's linear logic [3]. An object of the dialectica category Dial($\mathcal{C}$) of [4] is a triple (U, X, α). Here U is an object of witnesses, X an object of counterexamples, and α a relation between them. In the thesis the relation is a subobject of $U \times X$. Over sets such a subobject is the same thing as a map $U \times X \to \mathbf{2}$. This second reading suggests replacing $\mathbf{2}$ by any lineale, in the sense of [5], and so letting the relation take values in a richer structure of truth values.

Lineale-valued relations are what several applications ask for. Relations valued in the natural numbers are used in the multirelational treatment of Petri nets, in [6, 7] and in the recent work of Elena di Lavore, Wilmer Leal and the author [9]. For the Girard variant Gir($\mathcal{C}$) the construction over a lineale is carried out in [6] and again in [8]. Here we carry it out for Dial($\mathcal{C}$).

Section 2 sets out the Dialectica construction over a lineale. Section 3 shows that the resulting category is symmetric monoidal closed, with the adjunction between tensor and linear implication given by the adjunction of the lineale, applied pointwise. Sections 4 and 5 give finite products and weak coproducts. Section 6 gives the modality. When the base has free commutative monoids, collecting counterexamples into multisets and combining their values with the multiplication of the lineale gives a linear exponential comonad satisfying Girard's isomorphisms. Section 7 works out the examples of the natural numbers and the Lawvere quantale, and Section 8 observes that commutative groups, ordered by equality, are lineales, with the integers as example.

2 The Dialectica construction over a lineale

The construction asks two things of its setting: a base category $\mathcal{C}$ in which the witnesses and counterexamples live, and an object of truth values L in the base in which the relations take their values. The base must be cartesian closed for the internal hom, and have finite coproducts for the additive conjunction. The truth values must form a lineale L, so that the conditions on morphisms can be composed and the tensor has a right adjoint. Fix a cartesian closed category $\mathcal{C}$ and an object L of $\mathcal{C}$.

(C1) $\mathcal{C}$ is cartesian closed: it has finite products $U \times X$ and exponentials V^U.

(C2) $\mathcal{C}$ has an initial object 0 and binary coproducts $U + V$.

(L) L is a *lineale* in $\mathcal{C}$ [5]: an internal poset, given by a subobject $\sqsubseteq \rightarrowtail L \times L$ that is reflexive, transitive and antisymmetric, together with maps

$$\otimes \colon L \times L \to L, \qquad e \colon 1 \to L, \qquad \multimap \colon L \times L \to L$$

such that $(\otimes, e)$ is symmetric monoidal, $\otimes$ is monotone in each argument, and

$$(*) \qquad x \otimes y \sqsubseteq z \quad \text{if and only if} \quad x \sqsubseteq y \multimap z.$$

We argue with generalized elements, in the internal language of $\mathcal{C}$; an assertion $s \sqsubseteq t$ for $s, t \colon A \to L$ means that $\langle s, t \rangle \colon A \to L \times L$ factors through $\sqsubseteq$. Monotonicity and $(*)$ are assertions of this kind, so the arguments below are chains of internal implications and no external set theory enters. The

order of the lineale is still a subobject of $L \times L$: what the title gives up is subobjects as the relations of dialectica objects, not subobjects in the base. Asking for a lineale internal to $\mathcal{C}$ in this sense is a genuine assumption on the pair $(\mathcal{C}, L)$, and not a usual one in category theory; over sets any lineale will do, but in other bases one has to be found.

Definition 2.1. The category $\mathrm{Dial}_L(\mathcal{C})$ has as objects triples $A = (U, X, \alpha)$ with U, X objects of $\mathcal{C}$ and $\alpha\colon U \times X \to L$. A morphism $(U, X, \alpha) \to (V, Y, \beta)$ is a pair

$$f\colon U \to V, \qquad F\colon U \times Y \to X \qquad \text{with} \qquad \alpha\big(u, F(u,y)\big) \sqsubseteq \beta\big(f(u), y\big).$$

The identity on A is (id_U, π_2), and the composite of $(f, F)\colon A \to B$ and $(g, G)\colon B \to C$ is $g \circ f$ together with $(u, z) \mapsto F\big(u, G(f(u), z)\big)$.

With $L = \mathbf{2}$ over sets this is the category $\mathrm{Dial}_{\mathbf{2}}(\mathbf{Set})$ of [4]. Over sets the two readings of the relation, as a map into $\mathbf{2}$ and as a subobject, agree, but over other bases they need not. When the base is a poset, a Heyting algebra say, there is at most one map $U \times X \to L$, and the map-valued construction carries no information, while subobjects of $U \times X$ are the elements below it and remain plentiful; a Heyting algebra is the natural home for the lineale rather than for the base. The lineale being fixed throughout, we write $\mathrm{Dial}(\mathcal{C})$ for $\mathrm{Dial}_L(\mathcal{C})$ from now on.

Proposition 2.2. *Under* (C1) *and* (L)*,* $\mathrm{Dial}(\mathcal{C})$ *is a category.*

Proof. The identity condition is $\alpha(u, x) \sqsubseteq \alpha(u, x)$, which is reflexivity. For the composite, fix u and z and put $y = G(f(u), z)$. The hypothesis on (f, F) at (u, y) gives $\alpha(u, F(u, y)) \sqsubseteq \beta(f(u), y)$, and the hypothesis on (g, G) at $(f(u), z)$ gives $\beta(f(u), G(f(u), z)) \sqsubseteq \gamma(g(f(u)), z)$. The two right-hand and left-hand sides meet at $\beta(f(u), y)$, so transitivity gives the required inequality. Associativity and unitality hold because the forward maps compose in $\mathcal{C}$ and the backward maps are substitutions, associative and unital on the nose. □

Only finite products of $\mathcal{C}$ have been used so far. Exponentials enter with the internal hom, coproducts with the additive conjunction.

3 Monoidal closure

The tensor is taken coordinatewise: witnesses are paired with witnesses, counterexamples with counterexamples, and the two values are multiplied in the lineale.

Definition 3.1. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ put

$$\begin{aligned} A \otimes B &= \big(U \times V,\ X \times Y,\ \alpha \otimes \beta\big), \\ (\alpha \otimes \beta)\big((u,v),(x,y)\big) &= \alpha(u,x) \otimes \beta(v,y), \end{aligned}$$

with unit $I = (1, 1, e)$.

Lemma 3.2. *The tensor $\otimes$ is a bifunctor.*

Proof. Given $(f, F)\colon A \to A'$ and $(g, G)\colon B \to B'$, take $f \times g$ on witnesses and $F \times G$ on counterexamples, the latter reading $((u,v),(x',y')) \mapsto (F(u,x'), G(v,y'))$. The condition to check is

$$\alpha\big(u, F(u,x')\big) \otimes \beta\big(v, G(v,y')\big) \ \sqsubseteq\ \alpha'\big(f(u), x'\big) \otimes \beta'\big(g(v), y'\big),$$

which follows from the two hypotheses by monotonicity of $\otimes$ in each argument. Identities and composites are preserved because they are so in each coordinate. □

Monotonicity of $\otimes$ is used here and nowhere else; without it $\otimes$ is an operation on objects and not a functor.

Lemma 3.3. *The category* $(\mathrm{Dial}(\mathcal{C}), \otimes, I)$ *is symmetric monoidal.*

Proof. For the associator, take the associator of $\mathcal{C}$ on witnesses, $(U \times V) \times W \to U \times (V \times W)$, and the associator of $\mathcal{C}$ on counterexamples, $X \times (Y \times Z) \to (X \times Y) \times Z$, read backwards. Its condition is

$$\big(\alpha(u,x) \otimes \beta(v,y)\big) \otimes \gamma(w,z) \ \sqsubseteq\ \alpha(u,x) \otimes \big(\beta(v,y) \otimes \gamma(w,z)\big),$$

which is the associativity of $\otimes$ in L, as an equality, hence holds by reflexivity; the same maps in the other direction give the inverse. The left unitor has forward map $1 \times U \to U$ and backward map $X \to 1 \times X$, with condition $e \otimes \alpha(u,x) \sqsubseteq \alpha(u,x)$, the unit law of L. The symmetry is the symmetry of $\mathcal{C}$ in both coordinates, with condition the symmetry of $\otimes$. Each of the coherence diagrams commutes because it commutes in $\mathcal{C}$ in each of the two coordinates separately, the relation conditions having already been discharged. □

The linear implication is built from the morphisms themselves. Its witnesses are the data of a morphism, a forward map together with a backward map, and its counterexamples are the pairs at which such data might fail the morphism condition.

The witness object of $A \multimap B$ is the object of pairs $\langle f, F\rangle$ that a morphism $A \to B$ consists of, with the morphism condition itself as the relation: $\alpha(u, F(u,y)) \sqsubseteq \beta(f(u), y)$.

Definition 3.4. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ put

$$\begin{aligned} A \multimap B &= \big(V^U \times X^{U\times Y},\ U \times Y,\ \delta\big), \\ \delta\big(\langle f, F\rangle, (u, y)\big) &= \alpha\big(u, F(u, y)\big) \multimap \beta\big(f(u), y\big). \end{aligned}$$

The relation of $A \multimap B$ internalizes the morphism condition: a pair $\langle f, F\rangle$ satisfies the condition at (u, y) exactly when $e \sqsubseteq \delta(\langle f, F\rangle, (u, y))$, by $(*)$ and the unit law of the lineale.

Theorem 3.5. *Under* (C1) *and* (L)*, there is a bijection*

$$\mathrm{Dial}(\mathcal{C})\big(A \otimes B,\ C\big) \ \cong\ \mathrm{Dial}(\mathcal{C})\big(A,\ B \multimap C\big),$$

natural in A and C. Hence $\mathrm{Dial}(\mathcal{C})$ *is symmetric monoidal closed.*

Proof. Write $A = (U, X, \alpha)$, $B = (V, Y, \beta)$, $C = (W, Z, \gamma)$.

A morphism $A \otimes B \to C$ is a map $h\colon U \times V \to W$ together with $H\colon (U \times V) \times Z \to X \times Y$, and H is a pair $\langle H_1, H_2\rangle$ with $H_1\colon U \times V \times Z \to X$ and $H_2\colon U \times V \times Z \to Y$, subject to

$$(1) \qquad \alpha\big(u, H_1(u, v, z)\big) \otimes \beta\big(v, H_2(u, v, z)\big) \ \sqsubseteq\ \gamma\big(h(u, v), z\big).$$

A morphism $A \to B \multimap C$ is a map $k\colon U \to W^V \times Y^{V\times Z}$, that is a pair $\langle k_1, k_2\rangle$ with $k_1\colon U \to W^V$ and $k_2\colon U \to Y^{V\times Z}$, together with $K\colon U \times (V \times Z) \to X$, subject to

$$(2) \qquad \alpha\big(u, K(u, v, z)\big) \ \sqsubseteq\ \beta\big(v, k_2(u)(v, z)\big) \multimap \gamma\big(k_1(u)(v), z\big),$$

since the relation of $B \multimap C$ evaluated at $\langle k_1(u), k_2(u)\rangle$ and (v, z) is the right-hand side.

The data correspond: $h\colon U \times V \to W$ to $k_1\colon U \to W^V$ and $H_2\colon U \times V \times Z \to Y$ to $k_2\colon U \to Y^{V\times Z}$, both by exponential transpose in $\mathcal{C}$, and H_1 to K by the identity, both being maps $U \times V \times Z \to X$. Each is a bijection, so the underlying data of the two hom-sets are in bijection.

It remains to see that the two conditions correspond, and they do, because (1) is

$$\underbrace{\alpha(u, H_1)}_{x} \otimes \underbrace{\beta(v, H_2)}_{y} \ \sqsubseteq\ \underbrace{\gamma(h(u, v), z)}_{z'}$$

and (2) is $x \sqsubseteq y \multimap z'$ for the same three elements, under the correspondence just given. So (1) holds if and only if (2) does, by $(*)$.

For naturality in C, let $(g, G)\colon C \to C'$. Composing a morphism $A \otimes B \to C$ with it replaces h by $g \circ h$ and H by $(u, v, z') \mapsto H(u, v, G(h(u, v), z'))$;

composing the transposed morphism $A \to B \multimap C$ with $B \multimap (g, G)$ replaces k_1 by $u \mapsto g \circ k_1(u)$, k_2 by $u \mapsto k_2(u) \circ \langle \mathrm{id}, G(k_1(u)(-), -)\rangle$ and K correspondingly. These agree under the transposes above, since both are substitution of $G(h(u,v), z')$ for z and both leave the first coordinate to g. Naturality in A is the same computation on the other side, with $(f, F)\colon A' \to A$ substituting F into the counterexample coordinate. Functoriality of $\multimap$, contravariant in the first argument and covariant in the second, is then forced by the adjunction. □

The proof has two parts. Cartesian closure of the base rearranges the data: the exponential transposes match the forward and backward maps of a morphism $A \otimes B \to C$ with those of a morphism $A \to B \multimap C$. The adjunction $(*)$ of the lineale then matches the two conditions, moving $\beta(v, H_2)$ from the left of $\sqsubseteq$ to the right. The second part is the only one that sees the lineale, which is why the passage from **2** to a lineale costs nothing here.

4 Products

The additive conjunction pairs witnesses, as a conjunction should, but its counterexamples behave like a disjunction. To refute A and B it is enough to refute one of them, so a counterexample to the conjunction is a counterexample to A or a counterexample to B, and the challenger chooses which. The counterexample object is therefore the coproduct $X + Y$, and the relation consults whichever conjunct the counterexample belongs to.

Definition 4.1. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ put

$$A \times B \;=\; \big(U \times V,\; X + Y,\; \alpha \,\&\, \beta\big),$$

where the relation is given by cases,

$$\begin{aligned} (\alpha \,\&\, \beta)\big((u,v), \mathrm{inl}\, x\big) &= \alpha(u,x), \\ (\alpha \,\&\, \beta)\big((u,v), \mathrm{inr}\, y\big) &= \beta(v,y), \end{aligned}$$

that is, as the copairing of $\alpha \circ (\pi_1 \times \mathrm{id})$ and $\beta \circ (\pi_2 \times \mathrm{id})$ along the isomorphism $(U \times V) \times (X + Y) \cong (U \times V) \times X + (U \times V) \times Y$. Put $\top = (1, 0, !)$, with $!$ the unique map $1 \times 0 \to L$.

The isomorphism used is available because $(U \times V) \times (-)$ has a right adjoint by (C1) and so preserves coproducts; the initial object being the

empty coproduct, the same observation gives $U \times 0 \cong 0$. We use the latter in the form: a map out of $U \times 0$ is unique, and a condition indexed by $U \times 0$ holds.

Proposition 4.2. *Under* (C1), (C2) *and* (L), $A \times B$ *is the categorical product and* $\top$ *is terminal.*

Proof. Projections. Take $p_1 = (\pi_1, \mathrm{inl} \circ \pi_2)$, with forward map $\pi_1 \colon U \times V \to U$ and backward map $(u, v, x) \mapsto \mathrm{inl}\, x$. Its condition is $(\alpha \,\&\, \beta)((u, v), \mathrm{inl}\, x) \sqsubseteq \alpha(u, x)$, that is $\alpha(u, x) \sqsubseteq \alpha(u, x)$. Similarly for p_2.

Pairing. Given $(f, F) \colon C \to A$ and $(g, G) \colon C \to B$ with $C = (W, Z, \gamma)$, take $\langle f, g \rangle$ on witnesses and the copairing $[F, G] \colon W \times (X + Y) \to Z$ on counterexamples, using the isomorphism above. Its condition splits into the two cases

$$\gamma\big(w, F(w, x)\big) \sqsubseteq \alpha\big(f(w), x\big), \qquad \gamma\big(w, G(w, y)\big) \sqsubseteq \beta\big(g(w), y\big),$$

which are the hypotheses on (f, F) and on (g, G).

The triangles commute. The composite $p_1 \circ \langle (f, F), (g, G) \rangle$ has forward map $\pi_1 \circ \langle f, g \rangle = f$ and backward map $(w, x) \mapsto [F, G](w, \mathrm{inl}\, x) = F(w, x)$, so it is (f, F); likewise for p_2.

Uniqueness. A morphism $(t, T) \colon C \to A \times B$ with $p_1 \circ (t, T) = (f, F)$ and $p_2 \circ (t, T) = (g, G)$ has $\pi_1 \circ t = f$ and $\pi_2 \circ t = g$, so $t = \langle f, g \rangle$ by the universal property of the product in $\mathcal{C}$; and its backward map $T \colon W \times (X + Y) \to Z$ satisfies $T(w, \mathrm{inl}\, x) = F(w, x)$ and $T(w, \mathrm{inr}\, y) = G(w, y)$, so $T = [F, G]$ by the universal property of the coproduct in $\mathcal{C}$, transported along the isomorphism above.

Terminal. A morphism $A \to \top$ is a map $U \to 1$, unique, together with a map $U \times 0 \to X$, also unique since $U \times 0 \cong 0$; and the condition, being indexed by $U \times 0$, holds. So $\mathrm{Dial}(\mathcal{C})(A, \top)$ is a singleton. □

Proposition 4.3. *Every object of the form* $(0, X, \alpha)$ *is initial, and any two such are isomorphic.*

Proof. A morphism $(0, X, \alpha) \to (V, Y, \beta)$ is a map $0 \to V$, unique, together with a map $0 \times Y \to X$, also unique since $0 \times Y \cong 0$; and the condition, being indexed by $0 \times Y$, holds. So the hom-set is a singleton for every target. Between two objects $(0, X, \alpha)$ and $(0, X', \alpha')$ both hom-sets are then singletons, and each of the two composites lies in a singleton endo-hom-set, so each is the identity. □

The counterexample object carries no information once the witness object is 0, which is why initiality does not pin it down; this is the exact dual of the situation at $\top = (1, 0, !)$, where it is the witness object that is trivial and the counterexample object that is forced.

5 Weak coproducts

A witness for the additive disjunction is a witness for one of the two disjuncts, but a counterexample must be ready for either injection without being told which, so it is a pair of families of counterexamples, one indexed by each witness object.

Definition 5.1. For $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$ put

$$A \oplus B \;=\; \big(U + V,\; X^U \times Y^V,\; \alpha \oplus \beta\big),$$

where the relation is given by cases,

$$\begin{aligned}(\alpha \oplus \beta)\big(\operatorname{inl} u, (\varphi, \psi)\big) &= \alpha\big(u, \varphi(u)\big),\\ (\alpha \oplus \beta)\big(\operatorname{inr} v, (\varphi, \psi)\big) &= \beta\big(v, \psi(v)\big),\end{aligned}$$

that is, as a copairing along $(U+V) \times P \cong U \times P + V \times P$ for $P = X^U \times Y^V$, which holds because $P \times (-)$ is a left adjoint by (C1).

A witness $\operatorname{inl} u$ uses only the first family, at its own index, so the second component of the pair is untouched by the condition at $\operatorname{inl} u$, and that slack is the whole of the weakness.

Proposition 5.2. *Under* (C1), (C2) *and* (L) *there are morphisms* $i_1 \colon A \to A \oplus B$ *and* $i_2 \colon B \to A \oplus B$ *such that, for every* D *and every pair* $(f, F) \colon A \to D$, $(g, G) \colon B \to D$, *there is a morphism* $[\,(f, F), (g, G)\,] \colon A \oplus B \to D$ *with*

$$[\,(f, F), (g, G)\,] \circ i_1 = (f, F), \qquad [\,(f, F), (g, G)\,] \circ i_2 = (g, G).$$

Such a copairing need not be unique.

Proof. Injections. Take $i_1 = (\operatorname{inl}, \operatorname{ev}_U)$, with forward map $\operatorname{inl} \colon U \to U + V$ and backward map $U \times P \to X$ sending $(u, (\varphi, \psi))$ to $\varphi(u)$. Its condition is

$$\alpha\big(u, \varphi(u)\big) \;\sqsubseteq\; (\alpha \oplus \beta)\big(\operatorname{inl} u, (\varphi, \psi)\big),$$

an equality by Definition 5.1. Likewise $i_2 = (\operatorname{inr}, \operatorname{ev}_V)$.

Copairing. Let $D = (W, Z, \delta)$. Take $[f, g] \colon U + V \to W$ on witnesses, and on counterexamples the map

$$H \colon (U + V) \times Z \to X^U \times Y^V, \qquad H(w, z) \;=\; \big(\lambda u.\, F(u, z),\ \lambda v.\, G(v, z)\big),$$

which is the transpose of $\langle F, G\rangle$ precomposed with π_2 and so does not depend on w. Its condition splits along the injections. At $\operatorname{inl} u$ the left-hand side is $(\alpha \oplus \beta)(\operatorname{inl} u, H(\operatorname{inl} u, z))$, which by Definition 5.1 is the first component of H evaluated at u, that is $\alpha(u, F(u, z))$, and the required inequality $\alpha(u, F(u, z)) \sqsubseteq \delta(f(u), z)$ is the hypothesis on (f, F). At $\operatorname{inr} v$ it is $\beta(v, G(v, z)) \sqsubseteq \delta(g(v), z)$, the hypothesis on (g, G).

The triangles commute. The backward map of $[\,(f, F), (g, G)\,] \circ i_1$ sends (u, z) to $\mathrm{ev}_U(u, H(\operatorname{inl} u, z))$, the first component of H at u, which is $F(u, z)$; and the forward map is $[f, g] \circ \operatorname{inl} = f$. So the composite is (f, F), and likewise on the other side.

Failure of uniqueness. Write a backward map $H \colon (U+V)\times Z \to X^U \times Y^V$ as its two components $H_l \colon U \times Z \to X^U \times Y^V$ and $H_r \colon V \times Z \to X^U \times Y^V$. The condition at $\operatorname{inl} u$ mentions H_l only through the composite $(u, z) \mapsto \pi_1(H_l(u, z))(u)$, its first component evaluated at its own index, and the triangle with i_1 forces that composite to be F; dually the triangle with i_2 forces $(v, z) \mapsto \pi_2(H_r(v, z))(v)$ to be G. Nothing constrains $\pi_2 \circ H_l$ or $\pi_1 \circ H_r$, and nothing constrains $\pi_1 \circ H_l$ away from the index u. Any H agreeing with the one above in those two composites is therefore a copairing satisfying the same triangles, and over **Set** with U, V, X, Y each of at least two elements there are many. □

Exponentials are used here, so (C1) is needed for $\oplus$ and not only for $\multimap$; the coproduct of $\mathcal{C}$ is used for $U + V$ and for the copairing that defines the relation.

Remark 5.3. The weakness comes from the shape of the morphisms. The backward maps of $\mathrm{Dial}(\mathcal{C})$ receive the witness, so a counterexample for $A \oplus B$, produced before the side of the witness is known, must defer its answer, and the function spaces $X^U \times Y^V$ record that deferral. In $\mathrm{Gir}(\mathcal{C})$ backward maps do not see witnesses, and $(U + V, X \times Y)$ is a genuine coproduct. More generally, the morphisms of $\mathrm{Gir}(\mathcal{C})$ are symmetric, so swapping witnesses and counterexamples is a duality there, and coproducts and the par come from products and the tensor by duality. The morphisms of $\mathrm{Dial}(\mathcal{C})$ are not symmetric, and there is no such duality. This asymmetry is what places $\mathrm{Dial}(\mathcal{C})$ on the intuitionistic side.

Gödel's interpretation of disjunction makes a useful comparison. It takes witnesses $2 \times U \times V$ and counterexamples $X \times Y$, which requires the types to be inhabited, and there the slack lies in the default witnesses for the disjunct not chosen rather than in an unused family of counterexamples.

6 Modalities

Modelling the modalities requires one more hypothesis on the base.

(C3) $\mathcal{C}$ has pullbacks, and every object X has a free commutative monoid X^*, with unit $[\,]$, multiplication $\uplus$ and generators $\eta_X \colon X \to X^*$.

Free commutative monoids are a strong requirement for a cartesian closed category in general, but they exist in every topos with a natural numbers object and in every locally presentable cartesian closed category, so the usual bases satisfy (C3). Over sets X^* is the object of finite multisets on X, with $\eta_X(x) = [x]$. The lineale is itself a commutative monoid $(L, \otimes, e)$, and the modality collects counterexamples into a multiset and combines their values with the multiplication of the lineale. Two lemmas make this precise inside $\mathcal{C}$.

Lemma 6.1. *Let $(M, \cdot, 1)$ be a commutative monoid in $\mathcal{C}$ and $g \colon U \times X \to M$. Under* (C1) *and* (C3) *there is a unique $g^* \colon U \times X^* \to M$ that is a monoid map in its second argument, that is $g^*(u, [\,]) = 1$ and $g^*(u, \xi \uplus \xi') = g^*(u, \xi) \cdot g^*(u, \xi')$, and satisfies $g^*(u, \eta_X x) = g(u, x)$.*

Proof. The exponential M^U is a commutative monoid with the operations of M taken pointwise. Maps $U \times X^* \to M$ that are monoid maps in the second argument correspond, by transposition, to monoid maps $X^* \to M^U$, and the condition on generators corresponds to the condition that the monoid map extends the transpose $X \to M^U$ of g. Freeness of X^* gives existence and uniqueness of that extension. □

Over sets $g^*(u, \xi)$ is the product $\prod_{x \in \xi} g(u, x)$, with multiplicities. Applied to the relation of an object, with M the lineale, the lemma gives $\alpha^* \colon U \times X^* \to L$, and over sets $\alpha^*(u, \xi) = \bigotimes_{x \in \xi} \alpha(u, x)$. Uniqueness in Lemma 6.1 is used throughout in the following form: two maps $U \times X^* \to M$ that are monoid maps in the second argument and agree on generators are equal.

Lemma 6.2. *Let $g, h \colon U \times X \to L$ with $g(u, x) \sqsubseteq h(u, x)$. Under* (C1), (C3) *and* (L), *$g^*(u, \xi) \sqsubseteq h^*(u, \xi)$.*

Proof. First remove the parameter. Write $\sigma\colon U \times X^* \to (U \times X)^*$ for the extension, by Lemma 6.1, of $(u, x) \mapsto \eta(u, x)$, and $g^\natural\colon (U \times X)^* \to L$ for the monoid map extending g. The composite $g^\natural \circ \sigma$ is a monoid map in the second argument and sends $(u, \eta x)$ to $g(u, x)$, so it is g^* by uniqueness; likewise $h^* = h^\natural \circ \sigma$. It suffices to show $g^\natural \sqsubseteq h^\natural$.

Let $S \rightarrowtail (U \times X)^*$ be the pullback of $\sqsubseteq \rightarrowtail L \times L$ along $\langle g^\natural, h^\natural \rangle$. It contains $[\,]$, since $g^\natural[\,] = e = h^\natural[\,]$ and $\sqsubseteq$ is reflexive. It is closed under $\uplus$, since $g^\natural(\zeta \uplus \zeta') = g^\natural\zeta \otimes g^\natural\zeta'$ and similarly for $h^\natural$, and $\otimes$ is monotone in each argument. It contains the generators, since $g^\natural(\eta(u, x)) = g(u, x) \sqsubseteq h(u, x) = h^\natural(\eta(u, x))$. So S is a commutative monoid, the inclusion i is a monoid map, and η factors through S as η'. Freeness extends η' to a monoid map $k\colon (U \times X)^* \to S$, and $i \circ k$ is a monoid map extending η, hence the identity. A monic with a section is an isomorphism, so $\langle g^\natural, h^\natural \rangle$ factors through $\sqsubseteq$. □

Pullbacks are used here and nowhere else in the note.

Definition 6.3. For $A = (U, X, \alpha)$ put

$$!A \;=\; \big(U,\; X^*,\; \alpha^*\big).$$

For $(f, F)\colon A \to B$ with $B = (V, Y, \beta)$, the morphism $!(f, F)$ has forward map f and backward map $F^*\colon U \times Y^* \to X^*$, the extension of $\eta_X \circ F$ by Lemma 6.1.

A counterexample for $!A$ is a finite multiset of counterexamples for A, and a witness is measured against it by the product of the values. Over $\mathbf{2}$ the product is conjunction. The modality on $\mathrm{Gir}(\mathcal{C})$ is more complicated. There the backward maps do not see witnesses, so the internal hom is the simpler connective and the tensor needs function spaces in its counterexamples. The modality is tied to the tensor, since contraction lands in $!A \otimes !A$, and it inherits the difficulty: it is a composite of two comonads, one collecting counterexamples into multisets and one making them depend on witnesses, and composing them requires a distributive law between the two. In $\mathrm{Dial}(\mathcal{C})$ the tensor is coordinatewise and the modality is the first comonad alone.

Proposition 6.4. *Under* (C1), (C3) *and* (L), $!$ *is an endofunctor of* $\mathrm{Dial}(\mathcal{C})$.

Proof. The condition on $!(f, F)$ at (u, ξ) is

$$\alpha^*\big(u, F^*(u, \xi)\big) \;\sqsubseteq\; \beta^*\big(f(u), \xi\big).$$

The left-hand side is a monoid map in ξ, being a composite of two such, and on a generator ηy it is $\alpha(u, F(u, y))$; so it is g^* for $g(u, y) = \alpha(u, F(u, y))$.

The right-hand side is likewise h^* for $h(u,y) = \beta(f(u),y)$. The condition on (f,F) is $g \sqsubseteq h$, and Lemma 6.2 gives $g^* \sqsubseteq h^*$. Preservation of identities and composites is an equation between backward maps, and follows from uniqueness in Lemma 6.1: the extension of $\eta_X \circ \pi_2$ is π_2, and the extension of a composite is the composite of the extensions. □

Proposition 6.5. *Under* (C1), (C3) *and* (L), ! *is a comonad, with counit* $\varepsilon_A \colon !A \to A$ *and comultiplication* $\delta_A \colon !A \to !!A$ *given by the identity on witnesses and the backward maps*

$$\varepsilon_A \colon (u,x) \;\mapsto\; \eta_X x, \qquad \delta_A \colon (u,\Xi) \;\mapsto\; \mu_X \Xi,$$

where $\mu_X \colon X^{**} \to X^*$ *is the multiplication of the free commutative monoid monad.*

Proof. For ε_A the condition at (u,x) is $\alpha^*(u,\eta_X x) \sqsubseteq \alpha(u,x)$, an equality by Lemma 6.1.

For δ_A the condition at (u,Ξ) compares $\alpha^*(u,\mu_X\Xi)$ with $(\alpha^*)^*(u,\Xi)$. Both are monoid maps in Ξ, and on a generator $\eta\xi$ both are $\alpha^*(u,\xi)$, the first because $\mu_X \circ \eta_{X^*}$ is the identity. So they are equal, and the condition holds by reflexivity.

Naturality of ε and δ and the three comonad laws are equations between backward maps, the forward maps being identities throughout. They are the naturality, unit and associativity laws of the free commutative monoid monad; the lineale does not enter. □

Proposition 6.6. *Under* (C1), (C3) *and* (L), *each* $!A$ *is a commutative comonoid in* $(\mathrm{Dial}(\mathcal{C}), \otimes, I)$, *with weakening* $w_A \colon !A \to I$ *and contraction* $c_A \colon !A \to !A \otimes !A$ *given by*

$$w_A \;=\; \big(U \to 1,\ (u,*) \mapsto [\,]\big), \qquad c_A \;=\; \big(\Delta_U,\ (u,(\xi_1,\xi_2)) \mapsto \xi_1 \uplus \xi_2\big).$$

Proof. The condition on w_A is $\alpha^*(u,[\,]) \sqsubseteq e$, and $\alpha^*(u,[\,]) = e$. The condition on c_A at $(u,(\xi_1,\xi_2))$ is

$$\alpha^*\big(u, \xi_1 \uplus \xi_2\big) \;\sqsubseteq\; \alpha^*(u,\xi_1) \otimes \alpha^*(u,\xi_2),$$

an equality since $\alpha^*(u,-)$ is a monoid map. The comonoid laws are equations between maps of $\mathcal{C}$: on witnesses they are the laws of the diagonal, and on counterexamples the unit, associativity and commutativity of $\uplus$. □

Contraction holds with equality because the values are combined by $\otimes$. A modality built from the meet of the lineale instead would need $a \wedge b \sqsubseteq a \otimes b$ for contraction, which a general lineale does not satisfy.

Proposition 6.7. *Under* (C1), (C2), (C3) *and* (L) *there are Girard's isomorphisms*

$$!(A \times B) \;\cong\; !A \otimes !B, \qquad !\top \;\cong\; I.$$

Proof. In the commutative monoids of $\mathcal{C}$ finite products are also coproducts, and the free functor preserves coproducts, so the monoid map $k\colon (X+Y)^* \to X^* \times Y^*$ extending $\operatorname{inl} x \mapsto ([x],[\,])$ and $\operatorname{inr} y \mapsto ([\,],[y])$ is an isomorphism, and $0^* \cong 1$. Both objects of the first isomorphism have witness object $U \times V$, and the counterexample objects are related by k. Take the identity on witnesses and k, respectively k^{-1}, as backward maps, ignoring the witness. The relations correspond: the map

$$\zeta \;\mapsto\; \alpha^*\big(u, \pi_1 k\zeta\big) \otimes \beta^*\big(v, \pi_2 k\zeta\big)$$

is a monoid map in ζ, and on generators it sends $\operatorname{inl} x$ to $\alpha(u,x) \otimes e = \alpha(u,x)$ and $\operatorname{inr} y$ to $\beta(v,y)$, so it is $(\alpha \,\&\, \beta)^*((u,v),\zeta)$. Hence both conditions are equalities. For the second isomorphism, $!\top = (1, 0^*, !^*)$, and $!^*(*,[\,]) = e$, so $!\top \cong (1,1,e) = I$. □

Proposition 6.8. *Under* (C1), (C3) *and* (L), $!$ *is lax monoidal. The maps* $m_I\colon I \to !I$ *and* $m_{A,B}\colon !A \otimes !B \to !(A \otimes B)$ *are the identity on witnesses, with backward maps*

$$m_I\colon (*,\xi) \mapsto *, \qquad m_{A,B}\colon \big((u,v),\zeta\big) \;\mapsto\; \big(\pi_1^*\zeta,\ \pi_2^*\zeta\big),$$

where π_i^* *extends* $\eta \circ \pi_i$.

Proof. The condition on m_I is $e \sqsubseteq e^*(*,\xi)$. The constant map at e is a monoid map agreeing with e on generators, so $e^*(*,\xi) = e$.

The condition on $m_{A,B}$ at $((u,v),\zeta)$ is

$$\alpha^*\big(u, \pi_1^*\zeta\big) \otimes \beta^*\big(v, \pi_2^*\zeta\big) \;\sqsubseteq\; (\alpha \otimes \beta)^*\big((u,v),\zeta\big).$$

Both sides are monoid maps in ζ; the left because $\otimes$ is commutative and associative, so the pointwise product of two monoid maps into L is again one. On a generator $\eta(x,y)$ both are $\alpha(u,x) \otimes \beta(v,y)$. So they are equal. □

Theorem 6.9. *Under* (C1), (C3) *and* (L), $!$ *is a linear exponential comonad on* $\mathrm{Dial}(\mathcal{C})$ *in the sense of Bierman [10].*

Proof. Propositions 6.4, 6.5, 6.6 and 6.8 give the comonad, the comonoids and the lax monoidal structure, with every morphism condition checked.

The remaining requirements, that ε and δ are monoidal natural transformations, that w and c are monoidal natural transformations and morphisms of coalgebras, and that δ_A is a comonoid morphism, are equations between morphisms of $\mathrm{Dial}(\mathcal{C})$. A morphism is determined by its forward and backward maps, and none of these mentions L; the equations follow from the laws of the free commutative monoid monad and uniqueness in Lemma 6.1, exactly as over $\mathbf{2}$ in [4]. □

A category can carry more than one linear exponential comonad, so Theorem 6.9 alone does not determine the modality. Here it is determined: $!A$ is the cofree commutative comonoid on A, and so the only modality compatible with the comonoid structure of the tensor specified.

A commutative comonoid $C = (W, Z, \gamma)$ in $(\mathrm{Dial}(\mathcal{C}), \otimes, I)$ consists of a comultiplication $C \to C \otimes C$ and a counit $C \to I$. The counit laws force the forward map of the comultiplication to be the diagonal of W, so the comultiplication is given by a backward map $D\colon W \times (Z \times Z) \to Z$, and the counit by a backward map $o\colon W \to Z$, subject to

$$\gamma\big(w, D(w, z_1, z_2)\big) \;\sqsubseteq\; \gamma(w, z_1) \otimes \gamma(w, z_2), \qquad \gamma\big(w, o(w)\big) \;\sqsubseteq\; e.$$

The comonoid laws, read on backward maps, say that for each witness w the operation $D(w, -, -)$ with unit $o(w)$ is a commutative monoid structure on Z. The structure varies with w, so Z itself need not be a monoid in $\mathcal{C}$; but the exponential Z^W is one, with

$$(\varphi \cdot \varphi')(w) \;=\; D\big(w, \varphi(w), \varphi'(w)\big), \qquad 1(w) \;=\; o(w),$$

the laws holding pointwise. This is what replaces the fixed monoid M^U of Lemma 6.1.

Lemma 6.10. *Let $C = (W, Z, \gamma)$ be a commutative comonoid as above and $(f, F)\colon C \to A$ a morphism, with $A = (U, X, \alpha)$. Under* (C1)*,* (C3) *and* (L) *there is a unique $H\colon W \times X^* \to Z$ with*

$$H(w, [\,]) = o(w), \qquad H(w, \xi \uplus \xi') = D\big(w, H(w, \xi), H(w, \xi')\big), \qquad H(w, \eta_X x) = F(w, x),$$

and it satisfies $\gamma\big(w, H(w, \xi)\big) \sqsubseteq \alpha^*\big(f(w), \xi\big)$. Hence $(f, H)\colon C \to !A$ is a morphism of $\mathrm{Dial}(\mathcal{C})$.

Proof. The three equations say that the transpose $X^* \to Z^W$ of H is a monoid map extending the transpose $X \to Z^W$ of F, so existence and uniqueness of H are freeness of X^*.

For the inequality, consider the commutative monoid $M = Z^W \times L^W$, the second factor with the operations of L taken pointwise, and the map $K\colon X \to M$ sending x to $\big(\lambda w.\, F(w,x),\ \lambda w.\, \alpha(f(w),x)\big)$. Its extension to a monoid map $K^*\colon X^* \to M$ sends ξ to $\big(\lambda w.\, H(w,\xi),\ \lambda w.\, \alpha^*(f(w),\xi)\big)$, by uniqueness in each component. Let $T \rightarrowtail M$ be the pullback of $\sqsubseteq^W \rightarrowtail (L \times L)^W$ along the map sending (φ,ψ) to $\lambda w.\, \big(\gamma(w,\varphi(w)),\ \psi(w)\big)$; exponentiation preserves monics, being a right adjoint, so T is a subobject, consisting of the pairs with $\gamma(w,\varphi(w)) \sqsubseteq \psi(w)$ for all w. It is a submonoid. It contains the unit, by the counit condition $\gamma(w,o(w)) \sqsubseteq e$. It is closed under multiplication, since

$$\gamma\big(w, D(w,\varphi(w),\varphi'(w))\big) \ \sqsubseteq\ \gamma(w,\varphi(w)) \otimes \gamma(w,\varphi'(w)) \ \sqsubseteq\ \psi(w) \otimes \psi'(w)$$

by the comultiplication condition and monotonicity of $\otimes$. And K factors through T, which is the condition on (f,F). As in Lemma 6.2, freeness then makes K^* factor through T, and this is the required inequality. □

The comonoid structure of C enters only through the two conditions on its backward maps, and each is used once: the counit for the empty multiset and the comultiplication for $\uplus$. This is the one place where the lineale enters the cofreeness of !.

Proposition 6.11. *Under* (C1), (C3) *and* (L), $!A$ *is the cofree commutative comonoid on* A*: the functor* ! *is right adjoint to the forgetful functor from commutative comonoids in* $(\mathrm{Dial}(\mathcal{C}), \otimes, I)$ *to* $\mathrm{Dial}(\mathcal{C})$*, with counit* ε*.*

Proof. The transpose of $(f,F)\colon C \to A$ is the morphism (f,H) of Lemma 6.10. That it is a morphism of comonoids, that its composite with ε_A is (f,F), and that it is the only comonoid morphism with this property are equations between forward and backward maps; they do not mention the lineale, and hold as over **2** in [4], uniqueness coming from the uniqueness of H in Lemma 6.10. □

Theorem 6.9 and the earlier sections together make $\mathrm{Dial}(\mathcal{C})$ a model of intuitionistic linear logic. The tensor and linear implication come from Section 3, the additive conjunction from Section 4, and the modality from this section. The additive disjunction is interpreted by the weak coproducts of Section 5, which support the introduction and elimination of the additive disjunction and the expected triangle equations, though not every equation that genuine coproducts would provide. The model is a good one for two reasons. First, the modality is the cofree commutative comonoid on

A (Proposition 6.11). It is therefore determined by the rest of the structure rather than chosen, which few models of linear logic achieve. Second, the model is not a list of requirements but a construction from familiar ingredients: the base supplies products, exponentials and free commutative monoids, the lineale supplies the truth values, and the structure follows from these.

7 Dialectica multirelations

Relations valued in the natural numbers count rather than decide: the value $\alpha(u, x)$ says how often, or at what cost, the witness u meets the counterexample x. Two lineales of this kind, over the base **Set**, give the main examples. Both order numbers in reverse, so that smaller numbers are truer, the unit 0 is the top element, and the multiplication is addition. The base **Set** satisfies (C1)–(C3), so all the structure of the previous sections is available.

Example 7.1. The natural numbers form a lineale

$$\mathbb{N} \;=\; (\mathbb{N},\, \geq,\, +,\, 0), \qquad y \multimap z \;=\; z \mathbin{\dot{-}} y,$$

with truncated subtraction as the implication, since $x + y \geq z$ holds exactly when $x \geq z \mathbin{\dot{-}} y$.

The reverse order is forced. With the usual order there is no implication: when $y > z$ no x satisfies $x + y \leq z$, while $x = 0$ would satisfy $x \leq y \multimap z$ for any choice of $y \multimap z$.

An object of $\mathrm{Dial}_{\mathbb{N}}(\mathbf{Set})$ is a matrix of natural numbers indexed by witnesses and counterexamples, and a morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ satisfies $\alpha(u, F(u, y)) \geq \beta(f(u), y)$ for all u and y. The tensor adds the two counts, and the relation of the linear implication,

$$\delta\big(\langle f, F\rangle, (u, y)\big) \;=\; \beta\big(f(u), y\big) \mathbin{\dot{-}} \alpha\big(u, F(u, y)\big),$$

is the amount by which $\langle f, F\rangle$ fails the morphism condition at (u, y); it is 0, the unit, exactly where the condition holds. The additive conjunction reads off whichever count the counterexample asks for, and the modality adds counts along a multiset, with multiplicities:

$$!\alpha(u, \xi) \;=\; \sum_{x \in \xi} \alpha(u, x).$$

Example 7.2. The Lawvere quantale [12] is a lineale

$$\mathbb{R}^{\infty}_{\geq 0} \;=\; ([0,\infty],\, \geq,\, +,\, 0), \qquad y \multimap z \;=\; \max(z-y,\, 0),$$

with the conventions $y \multimap \infty = \infty$ for y finite and $\infty \multimap z = 0$ for every z, so that in particular $\infty \multimap \infty = 0$.

Here the values are distances or costs, and an object of $\mathrm{Dial}_{\mathbb{R}^{\infty}_{\geq 0}}(\mathbf{Set})$ is a generalized distance between witnesses and counterexamples. The tensor adds distances, the linear implication again measures by how much the morphism condition fails, and the modality takes the total cost of a multiset of counterexamples. The natural numbers sit inside $[0,\infty]$ closed under addition, truncated subtraction and 0, so $\mathrm{Dial}_{\mathbb{N}}(\mathbf{Set})$ is a full subcategory of $\mathrm{Dial}_{\mathbb{R}^{\infty}_{\geq 0}}(\mathbf{Set})$, closed under the tensor, the linear implication, the products and the modality.

8 Groups are lineales

A commutative group $(G, \cdot, 1)$, ordered by equality, is a lineale, with $\otimes$ the group operation, $e = 1$ and

$$y \multimap z \;=\; y^{-1} \cdot z,$$

since $x \cdot y = z$ holds exactly when $x = y^{-1} \cdot z$. Monotonicity of $\otimes$ is automatic for the discrete order. In a base $\mathcal{C}$ the same holds for a commutative group object G, with $\sqsubseteq$ the diagonal and the inverse supplying $\multimap$, so G satisfies (L) and all the results of the previous sections apply to $\mathrm{Dial}_G(\mathcal{C})$. What is particular to a group is that every morphism condition is an equation,

$$\alpha\big(u, F(u,y)\big) \;=\; \beta\big(f(u), y\big),$$

so that $\mathrm{Dial}_G(\mathcal{C})$ has the equations of the Chu construction with the morphisms of the Dialectica construction, whose backward maps read the witness.

Example 8.1. The integers form a lineale $\mathbb{Z} = (\mathbb{Z},\, =,\, +,\, 0)$, with $y \multimap z = z - y$.

An object of $\mathrm{Dial}_{\mathbb{Z}}(\mathbf{Set})$ is a matrix of integers indexed by witnesses and counterexamples, and a morphism (f, F) preserves the entries exactly: $\alpha(u, F(u,y)) = \beta(f(u), y)$ for all u and y. The tensor adds entries, and the relation of the linear implication,

$$\delta\big(\langle f, F\rangle, (u,y)\big) \;=\; \beta\big(f(u), y\big) - \alpha\big(u, F(u,y)\big),$$

is the signed defect of $\langle f, F\rangle$ at (u, y); it is 0, the unit, exactly where the condition holds. Compared with $\mathrm{Dial}_{\mathbb{N}}(\mathbf{Set})$, where truncated subtraction records only by how much the condition fails in one direction, the integers record the failure in both. The modality adds entries along a multiset, $!\alpha(u, \xi) = \sum_{x \in \xi} \alpha(u, x)$, as for $\mathbb{N}$. On objects with entries in $\mathbb{N}$ an equation implies the inequality $\geq$, so these objects and the morphisms of $\mathrm{Dial}_{\mathbb{Z}}(\mathbf{Set})$ between them form a subcategory of $\mathrm{Dial}_{\mathbb{N}}(\mathbf{Set})$, not full.

9 Conclusions

Taking the relation of a dialectica object as a map into a lineale, rather than as a subobject, the original Dialectica construction keeps all its structure. Over a cartesian closed base with finite coproducts, $\mathrm{Dial}(\mathcal{C})$ is symmetric monoidal closed, with finite products and weak coproducts. When the base also has free commutative monoids, it carries the cofree modality, which satisfies Girard's isomorphisms. In each case the conditions on morphisms reduce to the axioms of the lineale. For the adjunction between tensor and linear implication, the base rearranges the data and the adjunction of the lineale matches the conditions; for the modality, the counterexamples are combined by the multiplication of the lineale.

These calculations are needed as a reference by work that uses lineale-valued dialectica categories. The weakness of the coproducts remains somewhat mysterious. It disappears in the preorder of objects ordered by the existence of morphisms, which is the level at which the set-theoretic applications work [11], and Gödel's own interpretation of disjunction, which needs inhabited types, places the slack in default witnesses instead.

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